# A3-PS003-TC442

## NovaFlow: Bandwidth-Honest Federated Traffic Forecasting

*PS-003 · Federated Traffic-Flow Prediction with Communication-Cost Evaluation*



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# Problem Description

### Bandwidth
Roadside sensors stream raw data to the cloud around the clock over metered cellular links.

### Data concentration
Every junction's travel records pile up in one central store - one breach exposes the whole city.

### Non-IID traffic
A busy highway and a quiet ramp behave differently, so one shared model is pulled in different directions.

**Our question:** *when does federating a traffic model actually pay off - in accuracy, fairness and bytes?*

THE QUESTION NOBODY MEASURES

**≈24×**

more bytes for one 30-round FedAvg training (4.29 MB) than uploading the raw sensor counts it learns from (0.18 MB).

*Measured on real data · 4 junction-clients · 36 sensors*

# Background

**36**
sensors

**15 min**
sampling interval

**2,101**
time steps

**48**
raw columns per sensor

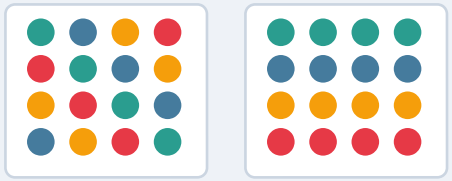
*UCI Traffic Flow Forecasting [7]: two highways, Northern Virginia / Washington D.C., CC BY 4.0*

## Federated learning

Junctions keep their data; only model weights travel. FedAvg averages client models by data size [1]:

$$w = \Sigma_k \, (n_k \, / \, n) \cdot w_k$$

## IID vs non-IID



IID non-IID

Rows = junctions. Non-IID: each junction owns its own roads, which slows and skews FedAvg [2].

## Why bytes matter

- Raw streaming cost grows with time × sensors.
- FL cost grows with model size × rounds × junctions.
- **Which is cheaper depends on the deployment - so it must be measured.**

# Why This Matters for Indian Cities

**ILLUSTRATIVE SCENARIO** One Smart City, four zones: a business district, an IT corridor, a residential belt and a highway bypass. Each zone trains on its own junction data; only model updates reach the traffic command centre.

### Mixed traffic is naturally non-IID

Two-wheelers, autos, buses and cycles share lanes, and the mix changes from zone to zone - exactly the client skew we test.

### Camera junctions flip the cost equation

Where junctions stream video, not counts, raw upload dwarfs model updates (one 2 Mbps camera ≈ 21.6 GB/day).

### Data stays inside each zone

Raw records never leave the zone - in the data-minimisation spirit of India's DPDP Act, 2023 [10] (no compliance claim).

### Honest about the data gap

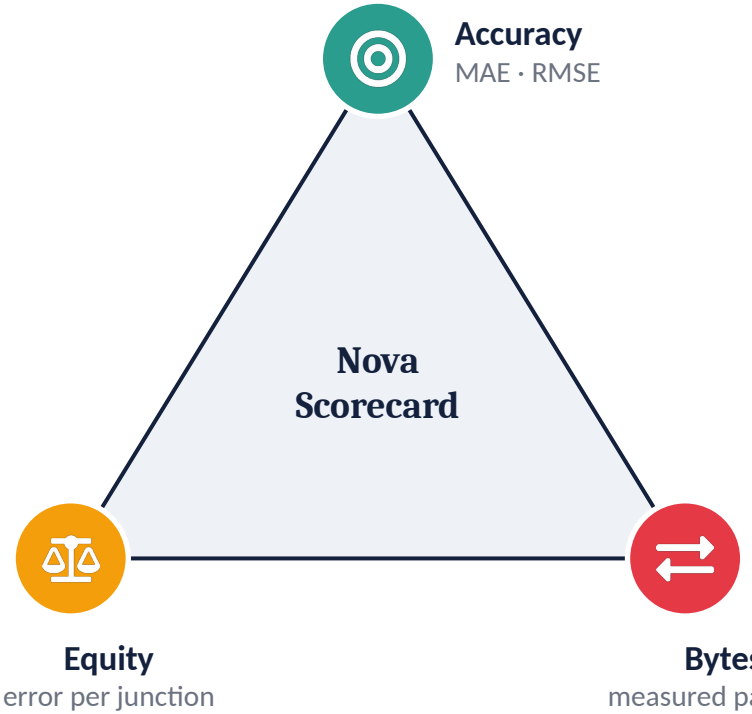
Open Indian traffic datasets are piecemeal [9], so we use a documented public benchmark; the simulator is dataset-independent.

# Literature & Standards Review

| Work | What it contributes | Gap NovaFlow addresses |
|---|---|---|
| **McMahan et al., 2017 [1]** | FedAvg: weighted averaging, far fewer rounds than FedSGD | Benchmarked on images/text, not traffic bytes |
| **Li et al., 2020 - FedProx [3]** | Proximal term stabilises heterogeneous clients | We test it on sensor-grouped (non-IID) traffic |
| **Konečný et al., 2016 [4]** | Quantised / sketched updates cut uplink cost | We measure real serialised bytes, both directions |
| **Liu et al., 2020 - FedGRU [5]** | FL for traffic-flow prediction | Reports accuracy, not a communication break-even |
| **Nishio & Yonetani, 2019 [8]** | Client selection by device resources | PulseFed selects by update size, audited for fairness |
| **IEEE Std 3652.1-2020 [6]** | Architectural framework for federated ML | We map our server / client / comms roles to it |

**Gap:** no study we found reports when FL actually pays off in bytes for scalar traffic sensors - while also checking fairness across junctions.

# NovaFlow: Measure Before You Federate




1. **Audit** verify time order and split before trusting the data
2. **Engineer** 30 features from the last 10 readings + calendar
3. **Partition** 4 junction-clients, IID and non-IID (quiet → busy)
4. **Train** centralised · FedAvg · PulseFed · FedProx · baselines
5. **Meter** serialise and count every byte, up and down
6. **Report** 3 seeds, auto-written report - no hand-typed numbers

# PulseFed: Talk Only When Learning Moves

**Illustrative 6-round schedule**




*16 uploads instead of FedAvg's 24 in this example - a sketch, not a result.*

1. **Score** after local training: r = ‖w_local − w_global‖ / ‖w_global‖
2. **Select** next round, clients whose model moved most talk; at least 2 of 4
3. **Refresh** no client sleeps more than 2 rounds in a row
4. **Aggregate** FedAvg over the active clients, weighted by samples

**Hypothesis:** fewer bytes per round for a small MAE cost.
**Guardrails:** saving capped near 50% (2 of 4 always talk); the Nova equity score checks that quiet roads are not silenced [8].

# Optimizations: Plain FedAvg vs NovaFlow

| Aspect | Before: plain FedAvg | After: NovaFlow |
|---|---|---|
| **Uplink per junction / round** | 17.8 KB float32 weights | **4.9 KB INT8 update (−73%)** |
| **Bytes per round (4 junctions)** | 142.6 KB | **90.8 KB (−36%)** |
| **Feature scaling** | Pool raw rows centrally | **Federated sums only (12.8 KB, once)** |
| **Non-IID drift** | Unhandled | **FedProx tested: no clear gain** |
| **Who talks each round** | All 4 junctions, always | **PulseFed: only movers (min 2)** |
| **Data leakage** | Assumed absent | **Audited by a time-order check** |

*Measured on the real dataset (mean of 3 seeds). Accuracy and PulseFed results: slides 11–13.*

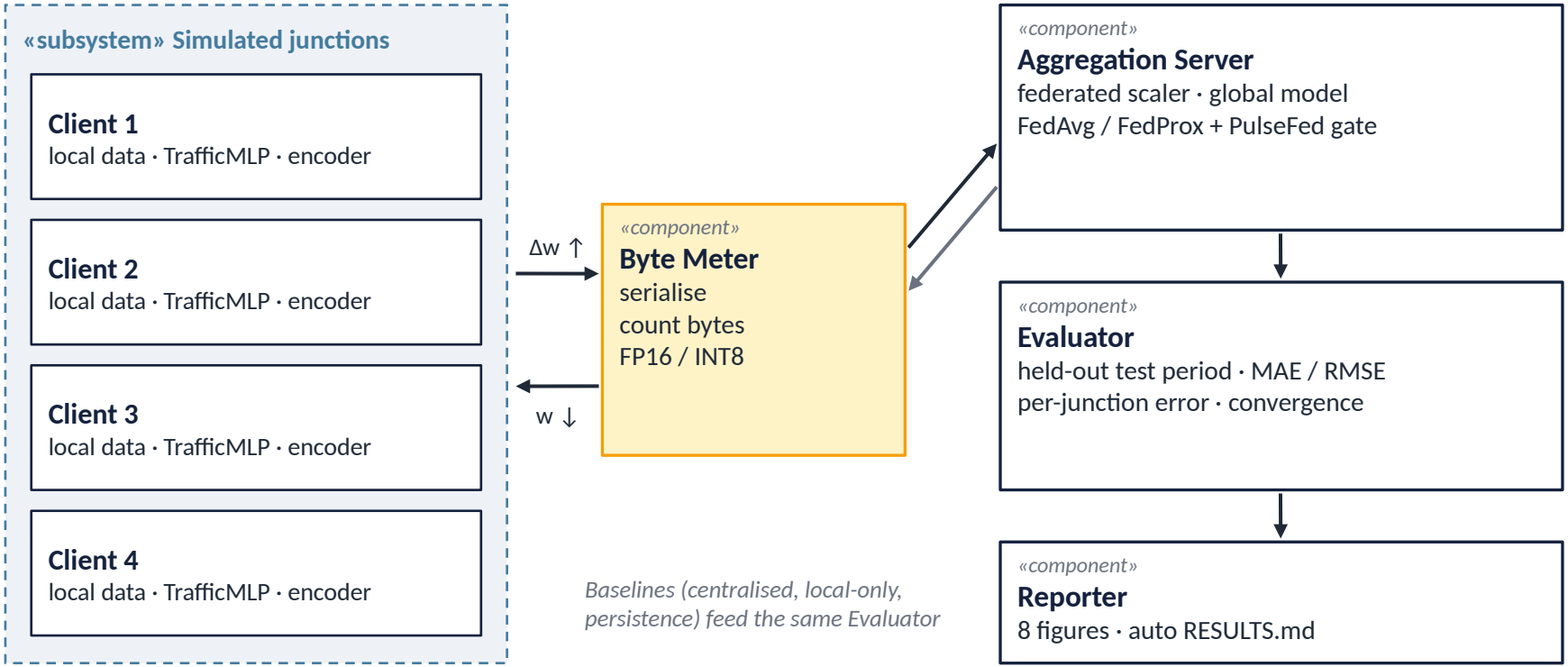
NEW METRIC: COMMUNICATION PAYBACK PERIOD

## ≈310 days

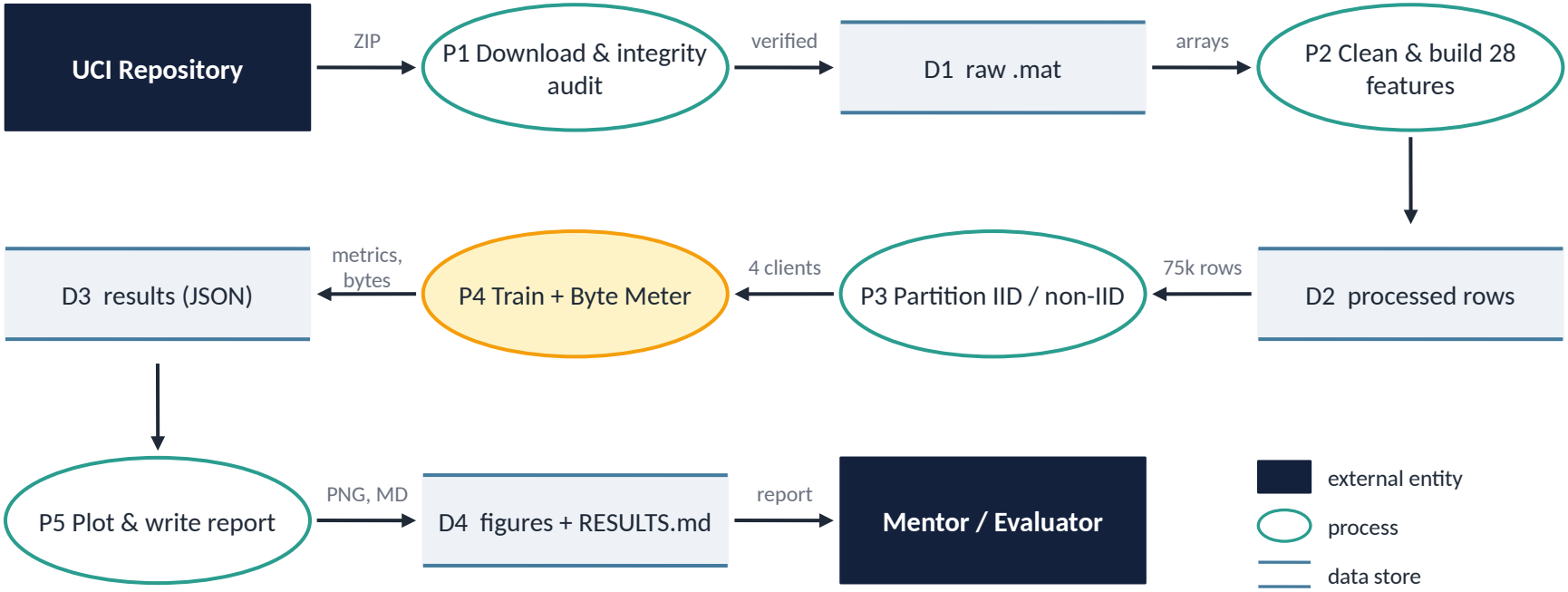
One 30-round training sends as many bytes as streaming all 36 sensors' raw counts for about 310 days.

*For scalar sensors FL must earn its keep through privacy and autonomy - or richer sensors: one 2 Mbps camera ≈ 21.6 GB/day.*

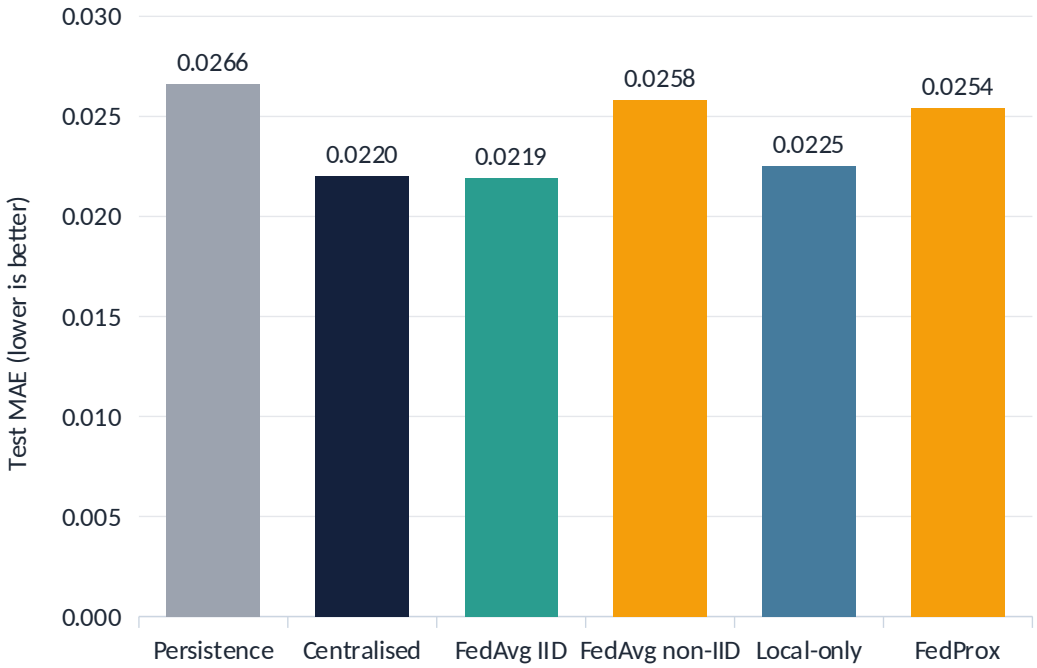
# Solution Architecture & Design



# Data Flow Diagram (Level 1)



# Results: Accuracy on Unseen Weeks




**−0.3%**

Similar junctions: FedAvg matches centralised, within 5% after 7–8 rounds.

**+17%**

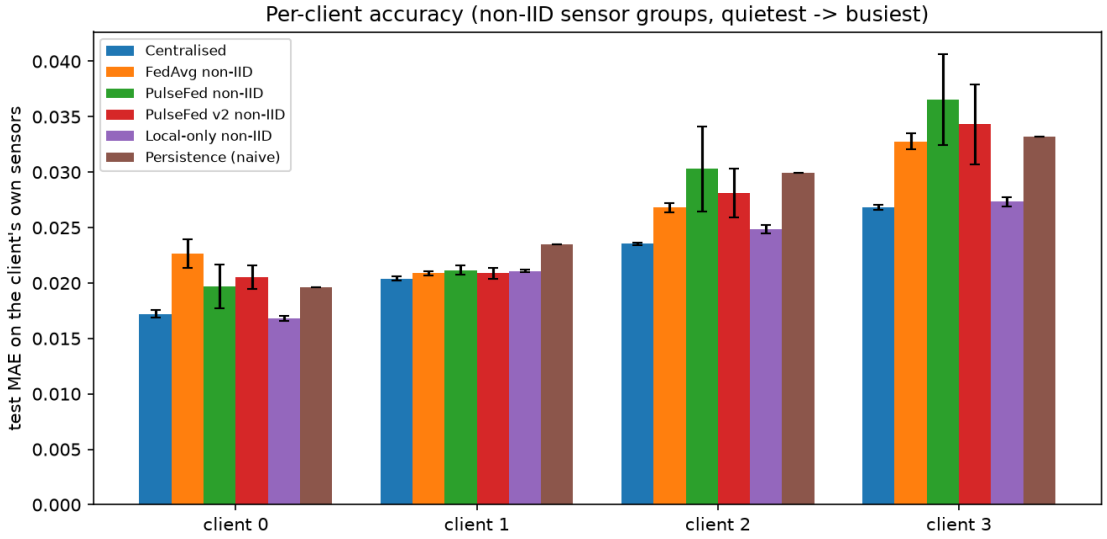
Different junctions (non-IID): error gap; never within 5% in 30 rounds.

**0.0225**

Local-only beats shared non-IID FedAvg (0.0258): personalised FL is next.

*Mean of 3 seeds · same held-out test weeks for every method · volumes scaled 0–1, so 0.022 ≈ 2.2% of peak traffic · full table: reports/RESULTS.md*

# Results: Who Pays the Price?




*Test MAE on each junction group's own roads, quietest → busiest (mean ± std, 3 seeds)*

**Quietest roads**

FedAvg is 31% worse than centralised, and even worse than the naive baseline (0.0226 vs 0.0196).

**Busiest roads**

FedAvg +22%. PulseFed v1 starved them (0.0365); v2 eased it (0.0343).

**Local-only**

Stays close to centralised on every road group.

**The overall MAE hides this.** One averaged model fits the 'average road'; our per-junction equity check exposed who loses.

# Results: Bytes vs Accuracy

| Method (non-IID) | Data sent | MAE | RMSE |
|---|---|---|---|
| **FedAvg** | 4.29 MB | 0.0258 | 0.0371 |
| **FedProx** | 4.29 MB | 0.0254 | 0.0366 |
| **FedAvg + FP16** | 3.17 MB (−26%) | 0.0257 | 0.0370 |
| **FedAvg + INT8 ★** | **2.74 MB (−36%)** | **0.0258** | **0.0371** |
| **PulseFed v1** | 2.76 MB (−36%) | 0.0269 | 0.0406 |
| **PulseFed v2** | 2.79 MB (−35%) | 0.0259 | 0.0378 |

**Payback period:** one FedAvg run (4.29 MB) ≈ 24× the raw counts ≈ 310 days of streaming all 36 sensors. For scalar sensors, FL's value is keeping data local.

**PULSEFED: WHAT WE LEARNED**

**v1:** −36% bytes, but MAE +4%, RMSE +9%; busiest roads starved.

**v2** (rule fixed before running): reuse sleeping junctions' last update. Average error back to FedAvg (0.0259 vs 0.0258).

*Busiest roads still hurt; with 3 seeds the gain is promising, not proven.*

**Verdict: INT8 gives the same saving, steady and fair. Our recommendation.**

# Deployment Timeline

**Oct 13–14**
Level 1: repo, this deck, intro video

**Oct 16–31**
Real-data runs + PulseFed, 3 seeds

**By Oct 31**
Code, report & demo link (Steps 9–10)

**Oct 31–Nov 4**
Evaluator connect & live demo

**Nov 24–30**
Regional hackathon

**Dec 17**
Finals, IIT Jammu

**Status:** Level 2 qualified; all 10 experiments × 3 seeds run on real data; 34 tests pass on every push.

**Test-driven**
34 automated tests

**Continuous checks**
tests run on every push

**Reproducible**
3 fixed seeds, mean ± std

**Zero hand-typed numbers**
report written from results

# Bibliography

[1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, B. Agüera y Arcas. “Communication-Efficient Learning of Deep Networks from Decentralized Data.” AISTATS 2017, PMLR 54, pp. 1273–1282.

[2] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, V. Chandra. “Federated Learning with Non-IID Data.” arXiv:1806.00582, 2018.

[3] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, V. Smith. “Federated Optimization in Heterogeneous Networks.” MLSys 2020.

[4] J. Konečný, H. B. McMahan, F. X. Yu, P. Richtárik, A. T. Suresh, D. Bacon. “Federated Learning: Strategies for Improving Communication Efficiency.” arXiv:1610.05492, 2016.

[5] Y. Liu, J. J. Q. Yu, J. Kang, D. Niyato, S. Zhang. “Privacy-Preserving Traffic Flow Prediction: A Federated Learning Approach.” IEEE Internet of Things Journal 7(8), 2020.

[6] IEEE Std 3652.1-2020. “IEEE Guide for Architectural Framework and Application of Federated Machine Learning.” IEEE, 2021.

[7] L. Zhao. “Traffic Flow Forecasting” [Dataset]. UCI Machine Learning Repository, 2019. doi:10.24432/C57897.

[8] T. Nishio, R. Yonetani. “Client Selection for Federated Learning with Heterogeneous Resources in Mobile Edge.” IEEE ICC 2019.

[9] IIT Madras, RBCDSAI. “Synchronized Multi Scale and Multi Sensor Traffic Data From Indian Urban Roads” (project page). rbcdsai.iitm.ac.in.

[10] Government of India. The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023).

[11] X. Gu, K. Huang, J. Zhang, L. Huang. “Fast Federated Learning in the Presence of Arbitrary Device Unavailability” (MIFA). NeurIPS 2021.

# Thank You for Your Attention!

*Measure before you federate.*



**Team Nova** · **A3-PS003-TC442**

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github.com/aiori-hackathon/A3-PS003-TC442